# Amal Mehta

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Research engineer with four years of applied deep learning across the stack: generative models and vision (GANs, CNN segmentation, detection) and deep RL and meta-RL, from data pipelines and multi-GPU training to policies deployed on physical robots. Prior neuroimaging research at UCSF; long-standing interest in how the brain learns.

## Experience

**Independent ML & Robotics Research** **Sept 2025 – Present**
*Self-directed projects; details at amalmehta.github.io/PortfolioWebsite* *San Francisco Bay Area*

- **Neural decoding:** 2D hand velocity from motor/premotor cortex spikes (NLB MC_Maze); GRU decoder outperformed a Kalman filter on held-out trials ($R^2$ 0.75 vs. 0.55).
- **Post-training:** SFT then GRPO on a small transformer policy in a key-and-door gridworld maze; over 3 seeds, GRPO lifted held-out success from 70% to 80% (DPO: 81%), where RL from scratch failed.
- **Self-play RL:** 9×9 Go engine trained from scratch with the AlphaGo Zero recipe; 7,040 self-play games in ∼11 h on a laptop.
- **Research tools:** Mac apps for 3D brain MRI (tissue segmentation, EEG placement) and for turning arXiv papers into runnable demos.
- **Pose estimation:** real-time sport technique analysis on MediaPipe 3D pose from a webcam; scores reps against one reference rep.
- Member, Sensorimotor AI journal club, Redwood Center (UC Berkeley); mentored undergraduate scholars from Berkeley.

**Deep Learning Scientist** **Sept 2021 – Aug 2025**
*PhysicsAI (DARPA-funded)* *Pacifica, CA (remote)*

- Formulated synthetic-data generation as RL: policies (TD3, PPO, DDPG) set image-generator parameters, rewarded by detector performance on real drone imagery, closing the sim-to-real gap without hand-tuned rendering.
- Built a PEARL meta-RL agent (probabilistic task embeddings over TD3) for fast few-shot adaptation across generator tasks.
- Implemented domain adaptation from scratch to hold detector performance in the low-real-label, high-synthetic-data regime.
- Built GAN-based synthesis pipelines (DCGAN, InfoGAN, FineGAN) producing training data for downstream detection models.

**Deep RL Researcher** **Aug 2023 – Mar 2024**
*Hybrid Robotics Lab, UC Berkeley — adv. Prof. Koushil Sreenath* *Berkeley, CA*

- Trained a single multi-stage PPO policy in IsaacGym for a Unitree Go1 quadruped to track, approach, and shoot a soccer ball at a target, replacing the multi-policy hierarchical framework used in prior work; ∼60% kicking accuracy on hardware.
- Designed the three-stage reward and transition machinery and CPG-based gait tracking; prototyped RLPD (SAC with prior data) to fine-tune the sim-trained policy on real robot data.

**Research Assistant** **June 2020 – Dec 2022**
*Video & Image Processing Lab, BAIR — adv. Prof. Avideh Zakhor* *UC Berkeley*

- Developed a deep CNN segmentation pipeline for invasive melanoma in large-scale histopathological images; published at SPIE Medical Imaging 2023; Stowell-Orbison Award Poster Session at USCAP 2022.

**Researcher** **June 2020 – Aug 2020**
*Abbasi-Asl Lab, UCSF — adv. Prof. Reza Abbasi-Asl* *San Francisco, CA*

- Built empirical mode decomposition (EMD) pipelines for 2D and 3D brain image registration.

## Education

**University of California, Berkeley** **Aug 2017 – Aug 2021**
*B.A. Computer Science — Regents Scholar* *Berkeley, CA*

- Focused on robotics and linear system theory; Regents, merit, and leadership awards.
- CS285 (Deep RL) project: iterative retraining of PPO agents against adversarial policies.

**Stanford University** **Aug 2021 – May 2023**
*Graduate coursework (non-degree) — Autonomous Systems & Artificial Intelligence*

- CS231N (Deep Learning for Vision), CS330 (Meta-Learning), AA271A/B (Autonomy I & II), AA273 (State Estimation).
- CS330 project: few-shot imitation learning; implemented $RL^2$ and inverse soft Q-learning in Meta-World.

## Technical Skills

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| **Languages** | Python, C++, C, Swift, TypeScript, JavaScript |
| **ML / RL** | PyTorch, TensorFlow, stable-baselines3, Gym, Meta-World, IsaacGym, OpenCV |
| **Product** | SwiftUI, AppKit, SceneKit, SwiftData, Core ML, Electron, Three.js, Next.js, React, SQLite |
| **Robotics** | ROS, Unitree Go1, CPG gait control, Kalman filtering |
| **Tools** | Git, GitHub Actions CI, Linux, Docker, CUDA/multi-GPU, W&B, TensorBoard, XCTest, Vitest, Playwright |

## Publications

1. Shah, A., **Mehta, A.**, Wang, M., Neumann, N., McCalmont, T., Zakhor, A. "Deep Learning Segmentation for Invasive Melanoma." *SPIE Medical Imaging*, 2023.
2. Neumann, N., Wang, M., **Mehta, A.**, et al. "Quantifying Invasive Melanoma Volume by Deep Learning Segmentation at the Pixel-Level." *USCAP Annual Meeting*, 2022.